CLINICAL · NECK, THYROID & SALIVARY

Salivary gland infections, stones and swellings

Most salivary swellings are not tumours. A child with painful, swollen cheeks has mumps; an old, dehydrated patient after surgery with a hot, tender parotid and pus at the duct has suppurative parotitis; a man whose submandibular gland swells every time he eats has a stone. Know the glands and their ducts, recognise the patterns of swelling (acute or recurrent, painful or painless, one gland or all), and remember that a persistent, firm lump in one gland is a tumour until proved otherwise.

AT A GLANCE

- Mealtime pain suggests a stone.
- Slow lump suggests tumour.
- Palsy with a lump suggests cancer.
- Ultrasound, FNA and careful surgery.
- Drooling: separate anterior from posterior; avoid duct relocation in children who aspirate.

1 CHAPTER ONE Applied anatomy

Three pairs of glands and hundreds of small ones.

1.1 Parotid gland

The largest, lying on the masseter and ramus, wrapped round the back of the mandible, from the zygomatic arch to below the angle (the tail), in front of the ear and sternomastoid; its deep surface reaches the prestyloid parapharyngeal space. It is enclosed in the parotidomasseteric fascia (part of the investing layer), whose unyielding nature makes parotitis so painful.

- **Facial nerve:** leaves the stylomastoid foramen, enters the gland and divides at the *pes anserinus* into temporofacial and cervicofacial divisions and five branches; it separates the larger *superficial lobe* (about 80%) from the *deep lobe*, the plane of parotid surgery. Landmarks for the trunk: the *tragal pointer* (the nerve lies about 1 cm deep and inferior to it), the *tympanomastoid suture*, and the *posterior belly of digastric* (the nerve is at its depth, above it).

- **Other contents:** the retromandibular vein (superficial to the external carotid), the external carotid artery dividing into maxillary and superficial temporal arteries, and intraparotid lymph nodes (about 20), the only salivary gland with intraglandular nodes.
- **Stensen's duct:** runs forward over the masseter, turns through the buccinator and opens opposite the upper second molar.
- **Nerves:** parasympathetic secretomotor fibres from the glossopharyngeal nerve via Jacobson's nerve, the lesser petrosal nerve, the otic ganglion and the auriculotemporal nerve; sensation of the overlying skin by the great auricular nerve (C2, C3).

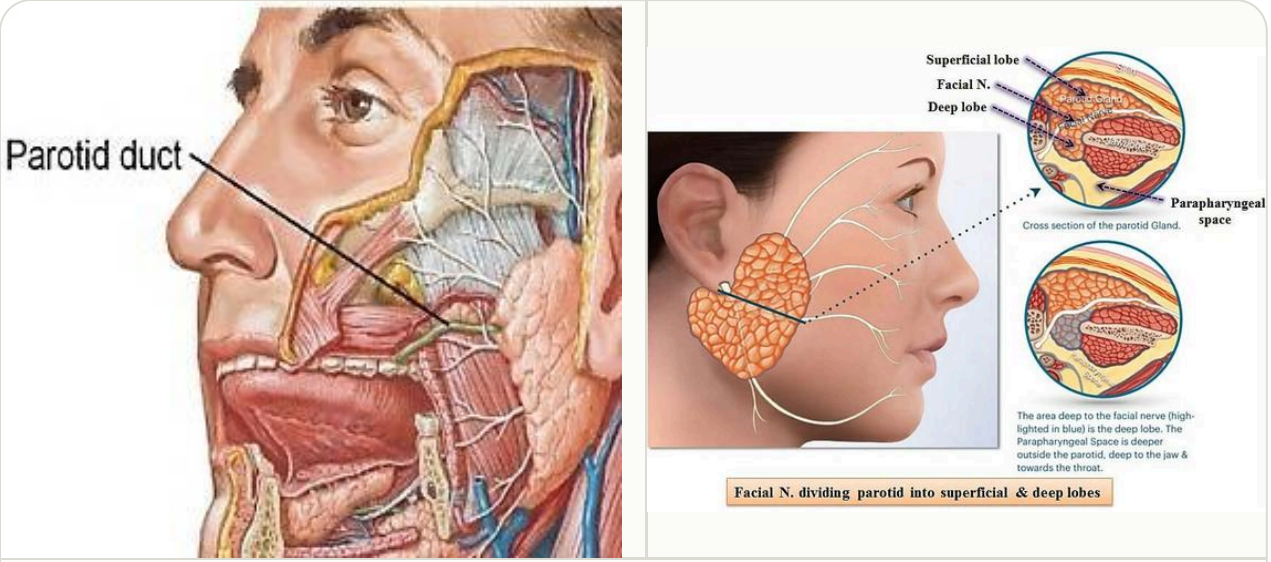
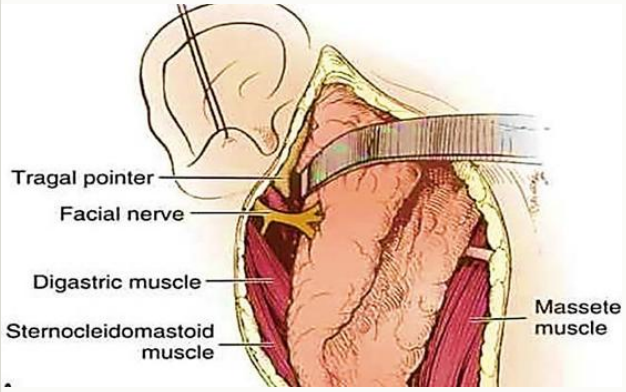


FIGURE 1 The parotid gland and Stensen's duct, and the facial nerve dividing it into superficial and deep lobes.

Anatomy atlas · ENT Steps p. 181



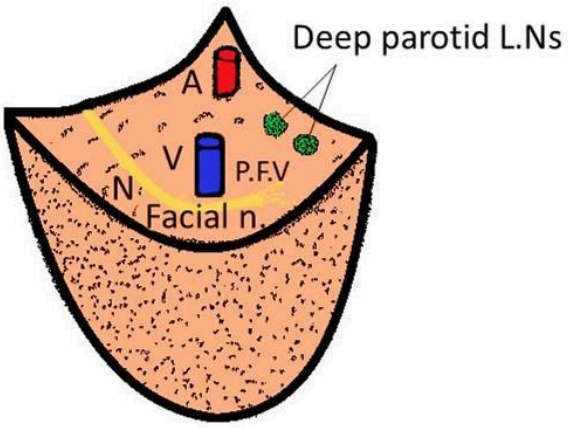
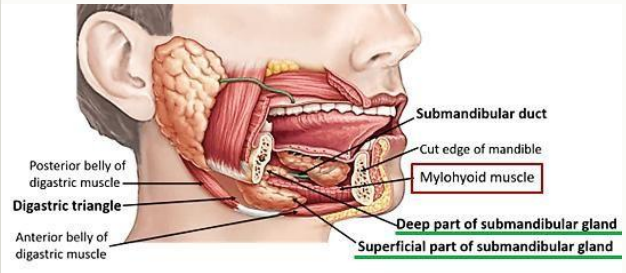


FIGURE 2 Landmarks for the facial nerve trunk, and the structures within the parotid.
Anatomy atlas · ENT Steps p. 185, Doss p. 74

1.2 Submandibular gland

In the digastric triangle, wrapped round the back of the mylohyoid: a large superficial part below it and a deep part above it. *Wharton's duct* runs forwards between the mylohyoid and hyoglossus, crossed by the *lingual nerve* (which loops beneath it), to open beside the frenulum. The facial artery grooves the deep part, the facial vein lies superficial to it, and the *marginal mandibular nerve* runs over the gland within its capsule; the hypoglossal nerve lies deep. Secretomotor supply: chorda tympani via the submandibular ganglion.



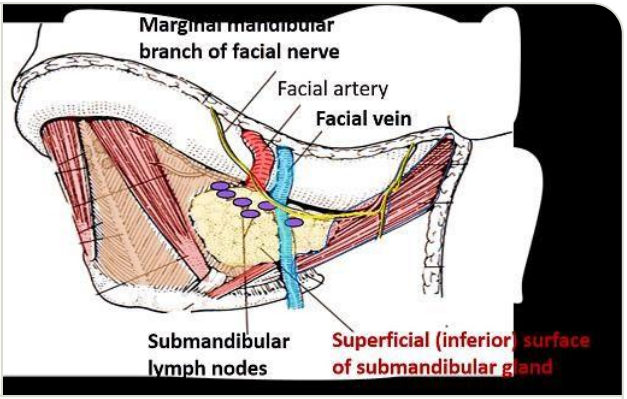


FIGURE 3 The submandibular gland around the mylohyoid, and the marginal mandibular nerve.
Anatomy atlas · ENT Steps p. 189, ENT Steps p. 190

1.3 Sublingual and minor glands

The sublingual glands lie under the floor of mouth mucosa above the mylohyoid, draining by many small ducts (of Rivinus), some into Wharton's duct (Bartholin's duct). Six hundred to a thousand minor salivary glands line the mouth, palate (most), pharynx, larynx and sinuses.

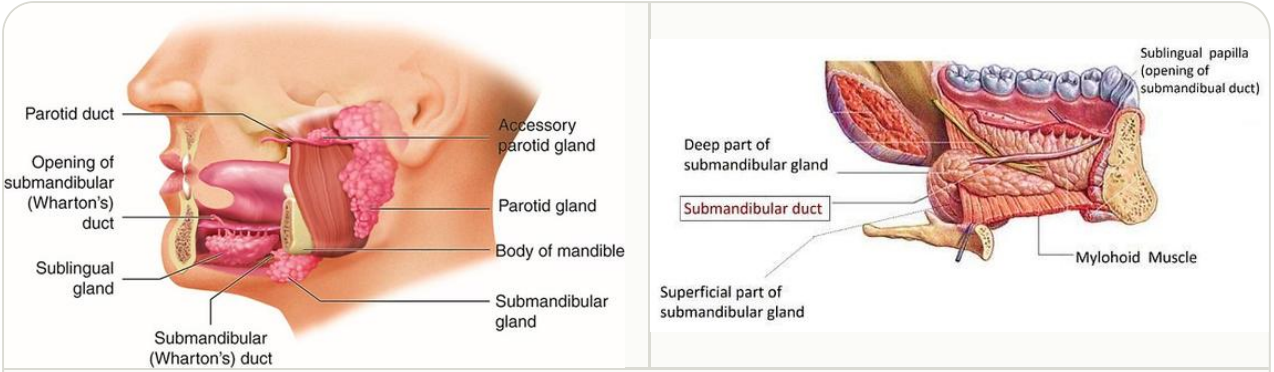


FIGURE 4 The major salivary glands and their ducts, and the sublingual and submandibular glands with their ducts.

Anatomy atlas · ENT Steps p. 176, ENT Steps p. 191

1.4 Saliva

About a litre to a litre and a half a day. At rest, most comes from the submandibular glands (more mucinous, richer in calcium); when stimulated by taste or smell, the parotids (serous, watery) supply most. Saliva lubricates, begins starch digestion (amylase), buffers acid (bicarbonate), protects teeth and kills microbes (IgA, lysozyme, lactoferrin). Flow is driven mainly by parasympathetic stimulation; dehydration, anticholinergic drugs, radiotherapy and age reduce it.

2 CHAPTER TWO Assessment

History, examination, imaging.

2.1 History and examination

Which gland, one or both sides; acute or recurrent; pain and swelling with meals (obstruction); fever and pus (infection); dry eyes and mouth (Sjögren); drugs,

alcohol, diabetes; and red flags for tumour (persistent mass, facial weakness, pain, skin fixation, nodes). Inspect and milk the duct orifices, and palpate *bimanually* the submandibular gland and floor of mouth (stones are often felt along the duct). Test the facial nerve.

2.2 Imaging

- **Ultrasound:** first line for stones, ductal dilatation, abscess and masses, and for guided aspiration.
- **Plain radiographs:** about 80% of submandibular stones (fewer parotid stones) are radio-opaque; an occlusal film shows duct stones.
- **Non-contrast CT:** detects stones reliably; contrast CT for abscess.
- **Sialography** (conventional or MR sialography): ducts, strictures, sialectasis and radiolucent stones; contraindicated in acute infection.
- **MRI:** the best study for masses.
- **Sialendoscopy:** both diagnosis and treatment.
- **Scintigraphy (technetium pertechnetate):** gland function; uptake in Warthin tumour and oncocytoma.

3 CHAPTER THREE Infections

Viral and bacterial.

3.1 Mumps (epidemic parotitis)

A paramyxovirus spread by droplets, saliva and urine; mostly children of four to six years in unvaccinated populations; incubation two to three weeks, infectious from before the swelling to about a week after.

- **Features:** fever, malaise and muscle pains, then painful swelling of one parotid and usually, within days, the other; sometimes the submandibular glands; swelling settles in about a week.
- **Complications:** orchitis (painful testis, usually one-sided; sterility rare), oophoritis, pancreatitis, aseptic meningitis and encephalitis, *unilateral sensorineural deafness*, sometimes sudden, thyroiditis, myocarditis, nephritis and arthritis.

- **Diagnosis:** clinical; raised serum and urinary amylase; IgM or rising IgG serology, or PCR.
- **Treatment:** supportive: fluids, rest, analgesics, warm or cold compresses, avoiding sour foods; support for orchitis.
- **Prevention:** MMR vaccine.

3.2 Acute suppurative sialadenitis

Usually the parotid, in the elderly, dehydrated and debilitated, after abdominal or hip surgery, and with a dry mouth from drugs: stasis of saliva lets mouth bacteria ascend Stensen's duct. *Staphylococcus aureus* (including MRSA), streptococci and anaerobes.

- Sudden, severe pain and swelling, worse on chewing, fever and toxicity; the duct orifice is red and swollen and pus can be milked from it.
- Treatment: rehydration, intravenous antibiotics covering staphylococci and anaerobes, sialogogues (lemon drops), massage from back to front, oral hygiene. Ultrasound or CT if it does not settle; an abscess (often hard to feel through the fascia) is drained through a parotidectomy incision with radial incisions in the line of the facial nerve branches.



FIGURE 5 Acute suppurative parotitis on the left in a 58-year-old woman.
Dhingra, fig. 45.1

3.3 Neonatal and childhood parotitis

Neonatal suppurative parotitis in premature babies (staphylococcal). *Juvenile recurrent parotitis*: recurrent swelling, often one-sided, in children aged three to ten without pus, with punctate sialectasis on imaging; usually remits at puberty; antibiotics during attacks, and sialendoscopic lavage with steroid.

3.4 Chronic recurrent sialadenitis and sialectasis

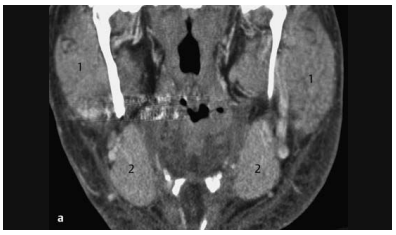
Recurrent painful swelling with pus during attacks and a firm, slightly enlarged gland between them, from duct obstruction, strictures, stones and reduced flow. Sialography shows ductal dilatation (punctate, globular or cavitary *sialectasis*) or strictures. Treat attacks with antibiotics; between them, hydration, oral hygiene, sialogogues and avoiding drying drugs; sialendoscopy to dilate strictures and wash out plugs; gland excision for persistent symptoms. The *Küttner tumour* (chronic sclerosing sialadenitis, part of IgG4-related disease) can mimic a submandibular neoplasm.

3.5 Granulomatous infections

- **Tuberculosis:** of the parotid (often its intraglandular nodes) or submandibular gland, as an acute sialadenitis or a chronic mass mimicking tumour, sometimes with skin changes or sinuses. Diagnosis by FNA (granulomas, acid-fast bacilli, culture) and tuberculin testing; antituberculous treatment.
- **Atypical mycobacteria:** in children of one to three years, a painless mass with a violaceous skin and sinus formation, a negative chest film and weak tuberculin reaction; excision or curettage.
- **Actinomycosis:** after dental infection; a woody swelling with sinuses discharging sulphur granules; prolonged penicillin.
- **Cat-scratch disease** (*Bartonella*) and **toxoplasmosis** cause periparotid and cervical lymphadenitis.

IMAGING IN THE APP 1 case

Sialadenitis on imaging.



Sialadenitis

Infection & inflammation · 11 images

4 CHAPTER FOUR Sialolithiasis

Stones.

4.1 Pathology

Calcium phosphate (hydroxyapatite) deposited on a nidus of mucus or debris. About 80 to 90% occur in the *submandibular* system, because its saliva is more alkaline, viscous and calcium-rich and its duct is long and runs uphill; about 10% in the parotid (often in the gland), rarely the sublingual.

4.2 Features

Pain and swelling of the gland at mealtimes (*salivary colic*), subsiding over an hour or two; recurrent infection with pus at the orifice; a stone may be seen at the papilla or felt along the duct bimanually.

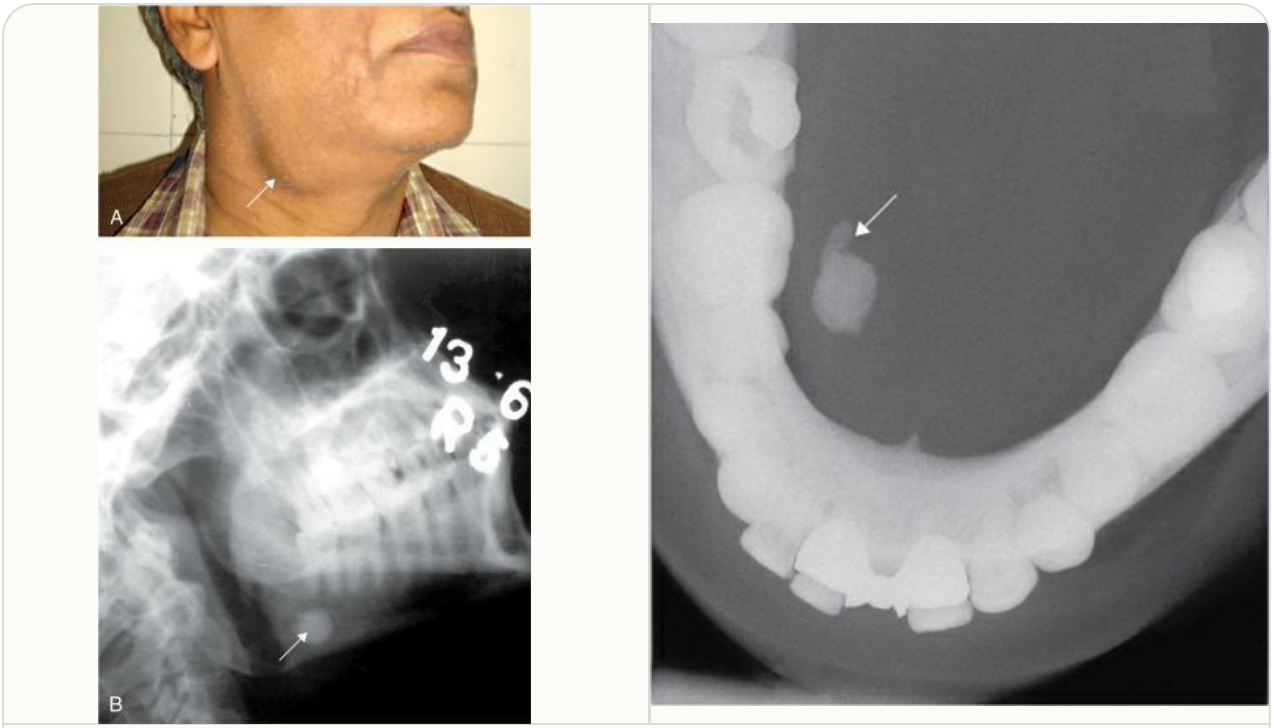


FIGURE 6 First, (A) an adult with submandibular sialadenitis (arrow) and (B) a radio-opaque stone in the submandibular gland on a lateral radiograph. Second, a radio-opaque stone in the right submandibular duct on an occlusal film.
Dhingra, figs 45.2 and 45.3

4.3 Diagnosis

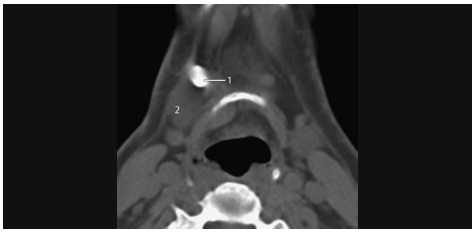
Plain and occlusal films (radio-opaque stones), ultrasound, non-contrast CT, and sialography or MR sialography for radiolucent stones and strictures.

4.4 Treatment

- **Small stones near the orifice:** conservative (hydration, sialogogues, massage) may pass them.
- **Stones in the anterior or middle duct:** intraoral *sialolithotomy*: a stay suture behind the stone to stop it slipping back, an incision over it along the duct, protecting the lingual nerve, and marsupialisation.
- **Sialendoscopy:** small stones (under about 5 mm) retrieved with a basket; larger ones fragmented by laser or extracorporeal lithotripsy; combined endoscopic and transoral approaches remove many hilar stones while keeping the gland.
- **Gland excision:** for stones in the hilum or parenchyma that cannot be reached, and for a gland destroyed by recurrent infection.

IMAGING IN THE APP 1 case

A submandibular sialolith on imaging.



Sialolith of the Submandibular Gland

Infection & inflammation · 5 images

OPERATIVE GUIDES IN THEATRE 2 operations

Sialendoscopy, and excision of the submandibular gland.

Salivary Endoscopy

Sialendoscopy

Excision of the Submandibular Gland

Submandibular gland

5 CHAPTER FIVE Systemic and autoimmune

Bilateral swelling and a dry mouth.

5.1 Sjögren syndrome

An autoimmune exocrinopathy with lymphocytic destruction of the salivary and lacrimal glands, mostly in women.

- **Primary:** dry mouth (xerostomia) and dry eyes (keratoconjunctivitis sicca), with parotid enlargement.
- **Secondary:** sicca features with another connective tissue disease, usually rheumatoid arthritis or lupus.
- **Diagnosis:** Schirmer test, anti-Ro (SS-A) and anti-La (SS-B) antibodies, rheumatoid factor and ANA, and *labial minor salivary gland biopsy* showing focal lymphocytic infiltrates; the gland shows a benign lymphoepithelial lesion (formerly Mikulicz disease).

- **Treatment:** saliva substitutes, frequent sips, sialogogues (pilocarpine, cevimeline), dental care, artificial tears.

LYMPHOMA IN SJÖGREN SYNDROME

The risk of non-Hodgkin lymphoma (usually MALT lymphoma of the parotid) is greatly increased. A new, firm or enlarging mass in a Sjögren gland is biopsied.

5.2 Sarcoidosis

Bilateral painless parotid swelling; *Heerfordt syndrome* (uveoparotid fever) combines parotitis, uveitis, fever and facial palsy. Steroids.

5.3 HIV

Bilateral parotid enlargement from *benign lymphoepithelial cysts* and lymphoid hyperplasia, sometimes the presenting sign of HIV; a sicca picture (diffuse infiltrative lymphocytosis syndrome). Antiretroviral treatment, aspiration or sclerotherapy for cosmetically troublesome cysts; a solid mass needs biopsy because of the risk of lymphoma.

5.4 Sialadenosis (sialosis)

Painless, symmetrical, non-inflammatory enlargement of the parotids from acinar hypertrophy, with diabetes, alcoholism, malnutrition, bulimia, obesity and drugs. Distinguish from other causes of bilateral swelling by FNA or biopsy; treat the cause.

5.5 Necrotising sialometaplasia

An ulcer or swelling of the hard palate (minor salivary glands) after ischaemic injury, mostly in men in their forties, whose pseudoepitheliomatous hyperplasia mimics squamous or mucoepidermoid carcinoma; it heals spontaneously in five to six weeks. Biopsy and wait.

CAUSES OF BILATERAL PAROTID SWELLING

GROUP	EXAMPLES
Infective	Mumps, HIV, recurrent parotitis of childhood
Autoimmune and granulomatous	Sjögren syndrome, sarcoidosis, IgG4-related disease
Metabolic	Sialadenosis (diabetes, alcohol, malnutrition, bulimia)
Neoplastic	Warthin tumour (bilateral in some), lymphoma, acinic cell carcinoma (rarely)
Drugs	Iodides, phenylbutazone and others

CHAPTER SIX

6 Dry and wet mouths

Xerostomia and sialorrhoea.

6.1 Xerostomia

From drugs (anticholinergics, antidepressants, antihistamines, diuretics), radiotherapy (irreversible damage above about 25 Gy), Sjögren syndrome, dehydration, diabetes and age. Effects: difficulty chewing, swallowing and speaking, caries, candida and altered taste. Treat with water, saliva substitutes, sugar-free gum, pilocarpine, fluoride and dental care; prevent radiation xerostomia with parotid-sparing IMRT and submandibular gland transfer.

6.2 Sialorrhoea (drooling)

Usually poor oral control rather than excess saliva, in children with cerebral palsy and adults with neurological disease. Positioning and therapy, anticholinergics (glycopyrrolate, hyoscine patches), botulinum toxin into the submandibular and parotid glands, and surgery (submandibular duct relocation or gland excision with parotid duct ligation).

DROOLING IN CHILDREN (OXFORD TEXTBOOK)

Drooling is normal up to about four years and usually stops by then in children without neuromuscular disease. Beyond that it is usually caused by poor oral and pharyngeal control, open lips and poor posture, as in cerebral palsy, rather than by too much saliva. At rest about 70% of saliva comes from the submandibular glands, which is why they are the main target of treatment. Separate **anterior drooling** (saliva spilling from the lips, a social and skin problem) from **posterior drooling** (saliva pooling in the pharynx and aspirated, causing cough, night choking and chest infections), because the choice of surgery depends on it. Severity and progress are measured with scales such as the Teacher's Drooling Scale or by counting bib changes.

Management is stepwise and multidisciplinary: oral motor training, posture and behavioural work; then **glycopyrronium** orally or a **hyoscine patch**; then ultrasound-guided **botulinum toxin** into the submandibular and parotid glands, which works for several months but can cause dysphagia if it spreads. Surgery is considered for persisting problems. **Duct ligation** (two, three or four ducts) is a simple day-case option but can recur. **Submandibular duct relocation** to the tonsillar fossa, often with sublingual gland excision to prevent a ranula, suits anterior drooling but can worsen aspiration and must be avoided in children who aspirate. **Bilateral submandibular gland excision**, with or without parotid duct ligation, is preferred for posterior drooling and aspiration.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Parotid anatomy:** the facial nerve and its landmarks; the course of Stensen's and Wharton's ducts.
- **Mumps** and its complications, including deafness.
- **Acute suppurative parotitis:** who gets it and why.
- **Why stones form in the submandibular gland,** and their management.
- **Sjögren syndrome:** diagnosis and the lymphoma risk.

- **Causes of bilateral parotid swelling.**

7.2 Common pitfalls

COMMON PITFALLS

- Performing sialography during acute infection.
- Calling a firm, persistent parotid lump "chronic parotitis" without imaging and FNA.
- Injuring the lingual nerve when removing a duct stone.

Written for this app from Oxford Textbook of Otolaryngology ch. 82, Oxford Specialist Handbook ch. 7, Oxford Handbook of ENT ch. 11, Dhingra ch. 45 and K. J. Lee ch. 35. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

SALIVARY GLANDS · INFECTION & INFLAMMATION

Sialadenitis

IMAGING FINDINGS

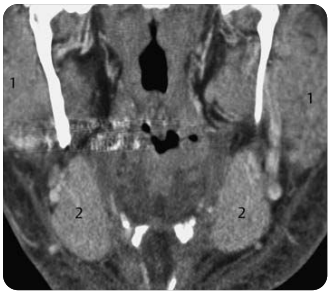
- Enlarged, enhancing gland with stranding of the surrounding fat
- Dilated duct, often with a stone
- Abscess: rim-enhancing collection

TOP DIFFERENTIALS

- Dental infection
- Nodal metastasis
- Gland tumour

CHECKLIST

- Look for a stone along the duct
- Chronic: small, fibrotic gland



Patient with chronic intermittent enlargement of the salivary glands and diffuse pain.
Dunnebier, Imaging for…



Axial CECT demonstrates an asymmetrically enhancing, enlarged left SMG compared to the right. Calculus is evident in the distal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 406.

SALIVARY GLANDS · INFECTION & INFLAMMATION

Sialolith of the Submandibular Gland

IMAGING FINDINGS

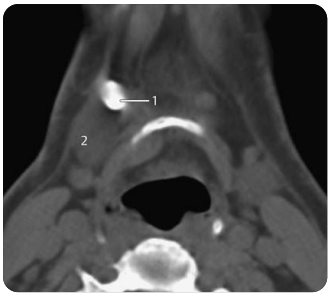
- Dense calculus in the Wharton duct or hilum on non-contrast CT
- Dilated duct proximal to the stone
- Enlarged inflamed or atrophic gland

TOP DIFFERENTIALS

- Phlebolith (venous malformation)
- Calcified node
- Dental fragment

CHECKLIST

- Position relative to the mylohyoid and hilum
- Ultrasound shows stone and duct without radiation



CT, axial bone window.A hyperdensity (1), with similar density as bone, suggestive of a sialolith (stone, calculus), probably obstructin outflo…



Axial CECT demonstrates an asymmetrically enhancing, enlarged left SMG compared to the right. Calculus is evident in the distal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 406.

SALIVARY GLANDS · INFECTION & INFLAMMATION

Ranula of the Sublingual Gland

IMAGING FINDINGS

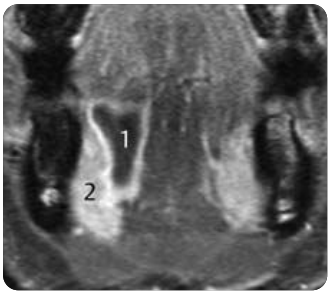
- Simple ranula: thin-walled cyst in the sublingual space
- Diving ranula: passes behind or through the mylohyoid into the submandibular space
- Tail sign pointing back to the sublingual space

TOP DIFFERENTIALS

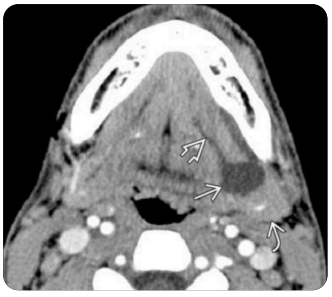
- Lymphatic malformation
- Dermoid or epidermoid
- Second branchial or thyroglossal cyst

CHECKLIST

- Relation to the mylohyoid decides the approach
- Remove the sublingual gland to prevent recurrence



MRI, T1-weighted with gadolinium enhancement and fat suppression, coronal. There is a cystic lesion suggestive of a ranula (1), originating…



Axial CECT demonstrates a rounded cystic mass in the submandibular space abutting the posterior margin of the mylohyoid muscle and displacing…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 400.